

COURSE CODE

SBT-DF202

COURSE TITLE

Computer and Digital Forensics

LAB TITLE

Digital Forensics Case Handling, Autopsy and Sleuth Kit
Analysis

NAME

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STUDENT ID

2025/FWSD/11325

DATE

29/08/2026

PLATFORM USED

Window Autopsy Software and Kali Linux

INSTRUCTOR

Mr. Aminu Idris, AMCPN
Commandant

Objective

By completing this lab, trainees will demonstrate their ability to apply the digital forensic investigation process in a controlled practical environment. Students will preserve and hash a forensic disk image, perform structured analysis using Autopsy, examine the image using The Sleuth Kit command-line tools, identify and recover deleted files, analyse file metadata and data units, perform keyword searches, verify recovered evidence using cryptographic hashes, and prepare a professional forensic report containing repeatable findings.

Check hash code online

File Hash Online Calculator^{WASM}

- Calculates MD5, SHA1, SHA2 (SHA256), and SHA512 hashes all at once
- The browser performs all calculations without uploading data to the server
- Supports unlimited files of any size

Drop files here or click to select

and hash them all

Choose Files Ch01InChap01.dd

Ch01InChap01.dd - 1474560 bytes

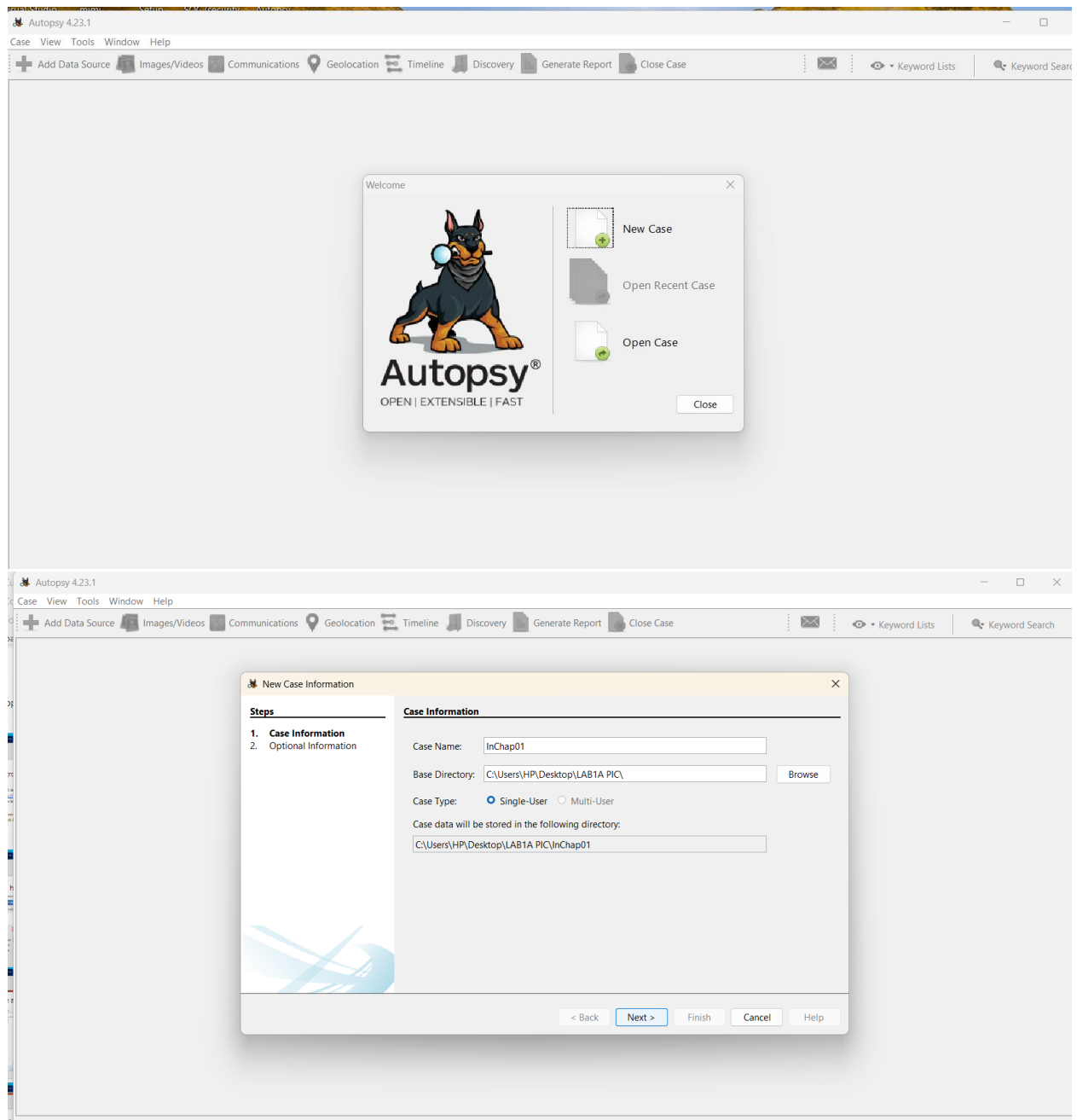
MD5: a117773bcf1fc88ec0ab8e0a349fbbcb

SHA1: 6200453567102ec6100adfa91cdbf7b1003574f6

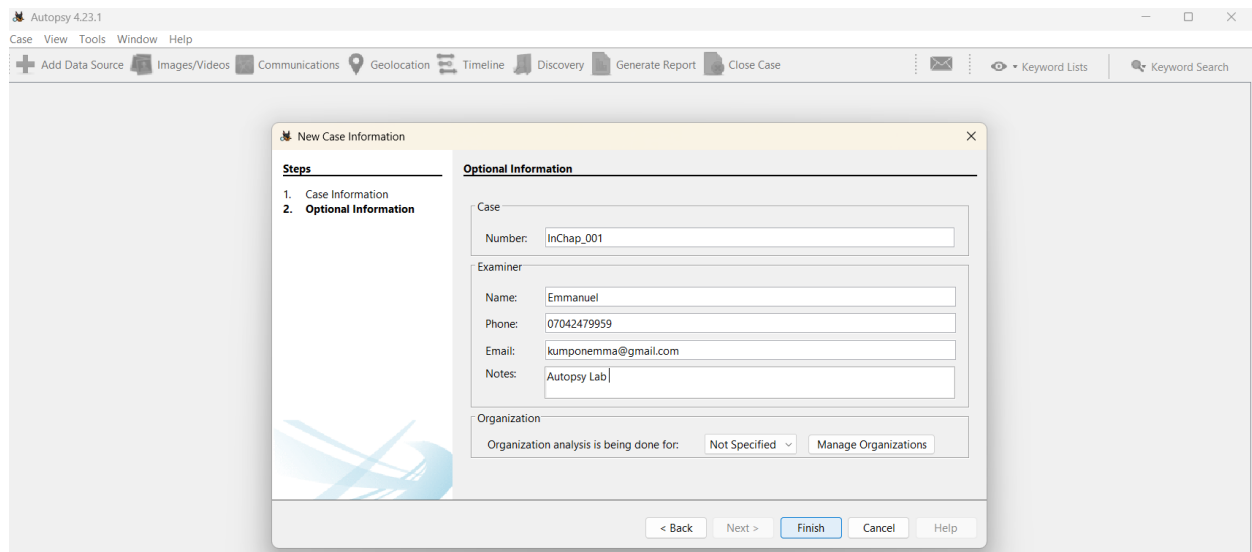
SHA256: 3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

SHA512: 0c02f8a8388e81664da0f2c46d97fccfa6dd3d4a7355a60fda3a3c9c282bda7149d1846f2853537b02184092e996ea28df1c90a1b78f9b8f1ce0ce7c700f25bc

Create a case with name



Details of the case



The image shows the 'New Case Information' dialog box in Autopsy 4.23.1. The dialog has a 'Steps' panel on the left with two steps: '1. Case Information' and '2. Optional Information'. The 'Optional Information' step is currently selected. The main area contains several input fields: 'Case Number' (filled with 'InChap_001'), 'Examiner Name' (filled with 'Emmanuel'), 'Phone' (filled with '07042479959'), 'Email' (filled with 'kumponemma@gmail.com'), and 'Notes' (filled with 'Autopsy Lab'). There is also an 'Organization' section with a dropdown menu set to 'Not Specified' and a 'Manage Organizations' button. At the bottom, there are buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

Autopsy 4.23.1

Case View Tools Window Help

+ Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

New Case Information

Steps

1. Case Information
2. **Optional Information**

Optional Information

Case

Number: InChap_001

Examiner

Name: Emmanuel

Phone: 07042479959

Email: kumponemma@gmail.com

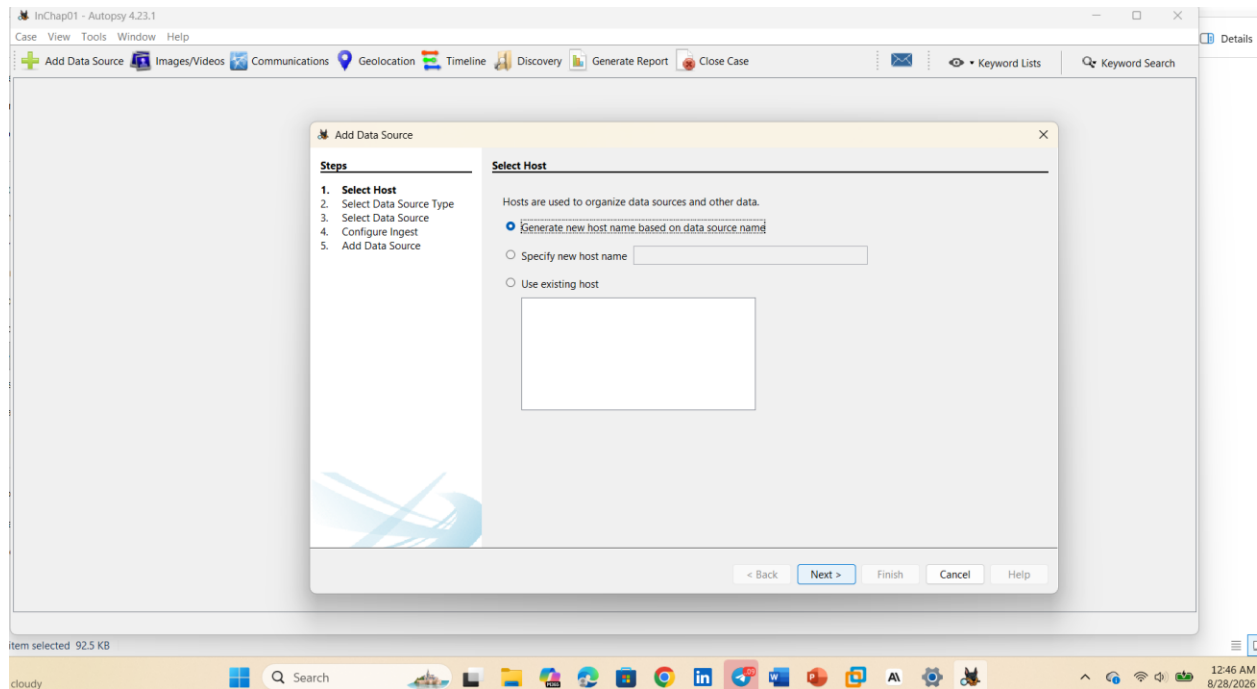
Notes: Autopsy Lab

Organization

Organization analysis is being done for: Not Specified Manage Organizations

< Back Next > Finish Cancel Help

Choose Date Format



The image shows the 'Add Data Source' dialog box in Autopsy 4.23.1. The dialog has a 'Steps' panel on the left with five steps: '1. Select Host', '2. Select Data Source Type', '3. Select Data Source', '4. Configure Ingest', and '5. Add Data Source'. The 'Select Host' step is currently selected. The main area contains a text box with the instruction 'Hosts are used to organize data sources and other data.' Below this, there are three radio buttons: 'Generate new host name based on data source name' (selected), 'Specify new host name' (with an empty text field), and 'Use existing host' (with an empty text field). At the bottom, there are buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

InChap01 - Autopsy 4.23.1

Case View Tools Window Help

+ Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Add Data Source

Steps

1. **Select Host**
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source

Select Host

Hosts are used to organize data sources and other data.

☒ Generate new host name based on data source name

☐ Specify new host name

☐ Use existing host

< Back Next > Finish Cancel Help

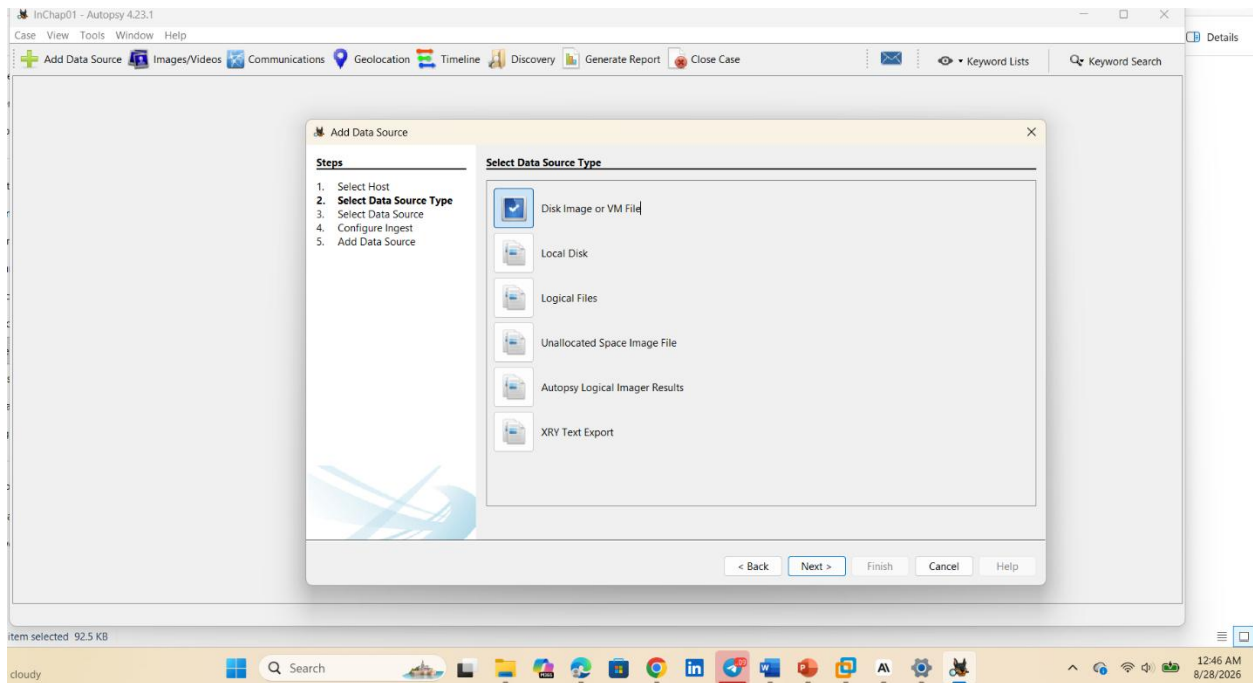
Item selected 92.5 KB

cloudy

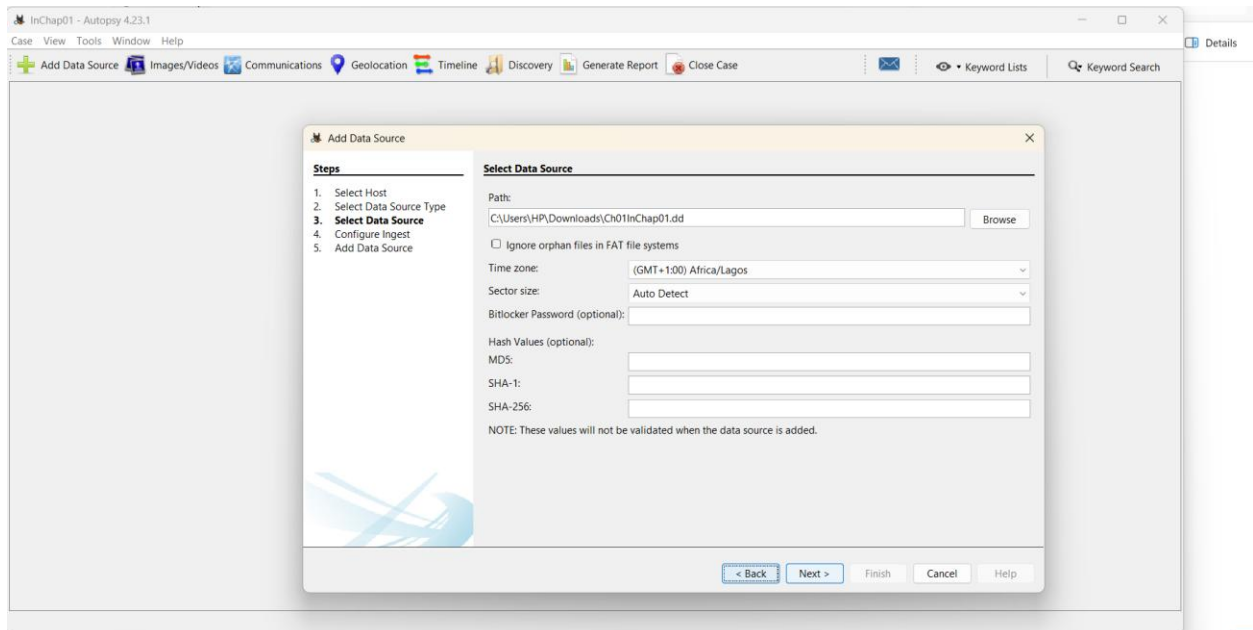
Search

12:46 AM 8/28/2026

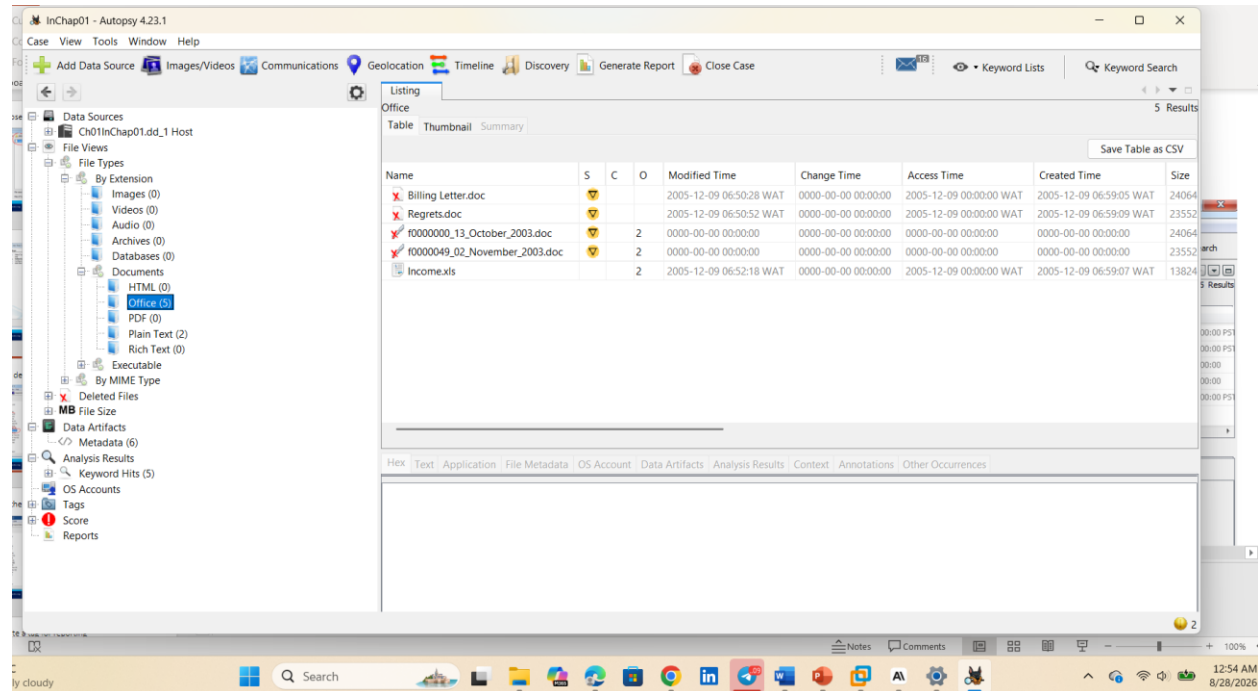
Click Disk Image or VM File



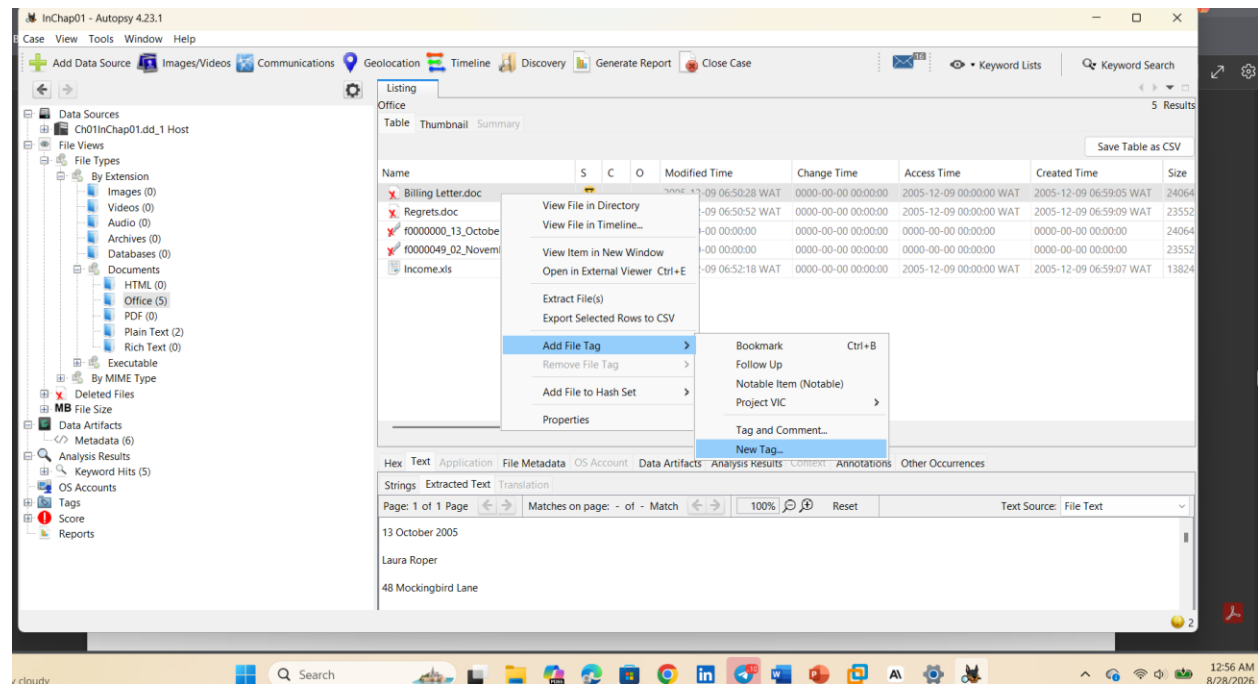
Choose the image file



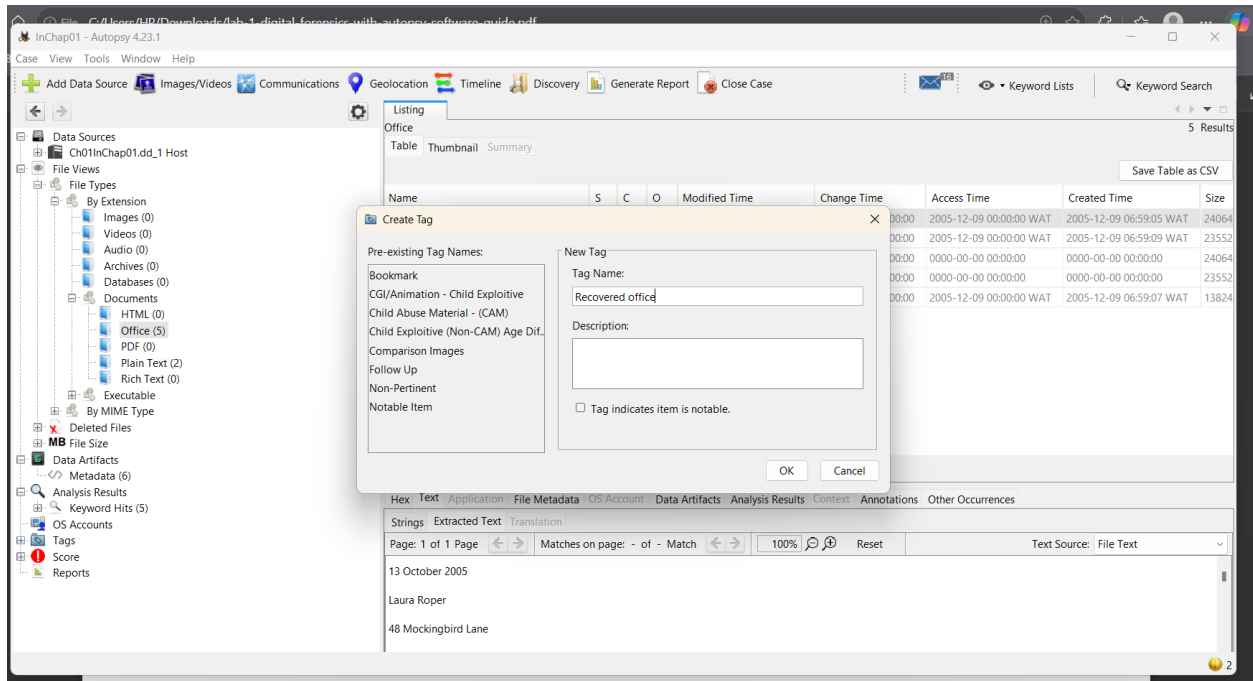
Find deleted files



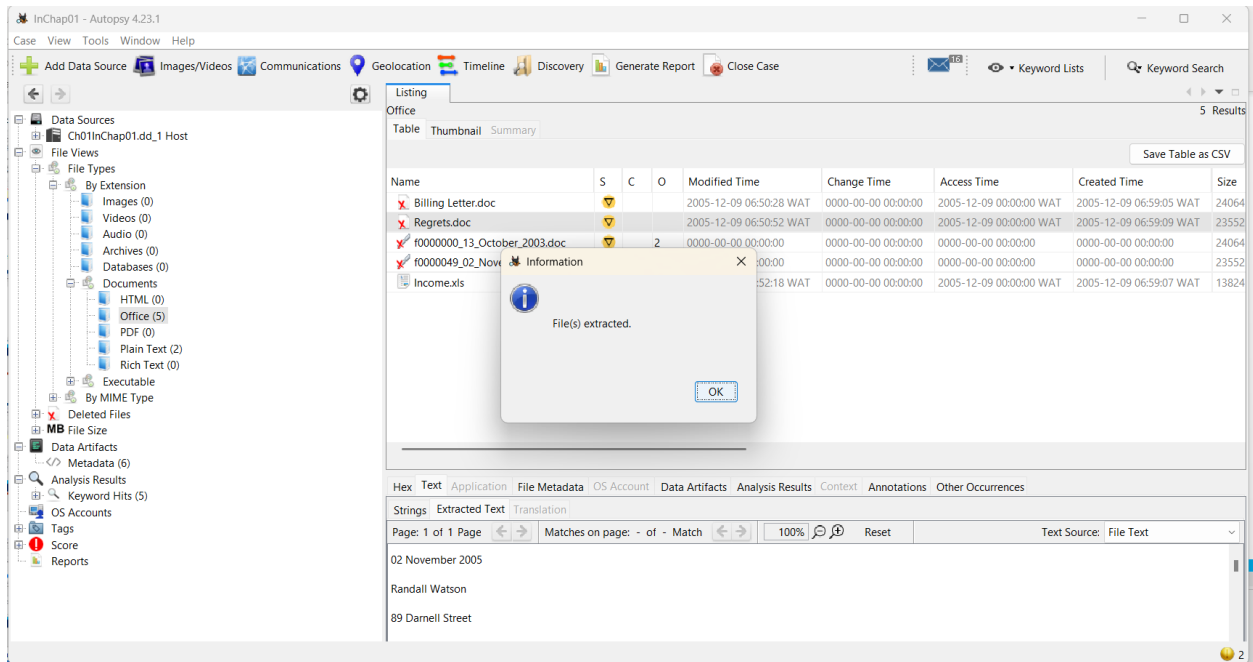
Tag the file



Create a tag reporting



Recover deleted file



Search keywords “George”

InChap01 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing Office

Table Thumbnail Summary

Name S C O Modified Time Change Time

Billing Letter.doc 2 2005-12-09 06:50:28 WAT 0000-00-00 00:00:00

Regrets.doc 2 2005-12-09 06:50:52 WAT 0000-00-00 00:00:00

r0000000_13_October_2003.doc 2 0000-00-00 00:00:00 0000-00-00 00:00:00

r0000049_02_November_2003.doc 2 0000-00-00 00:00:00 0000-00-00 00:00:00

Income.xls 2 2005-12-09 06:52:18 WAT 0000-00-00 00:00:00

George

Exact Match Substring Match Regular Expression

Restrict search to the selected data sources:

Ch01InChap01.dd

Save search results

Search Files Indexed: 0

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of 1 Page Matches on page: - of - Match 100% Reset Text Source: File Text

02 November 2005

Randall Watson

89 Darrell Street

InChap01 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Listing Keyword search 1 - George x

Keyword search 9 Results

Table Thumbnail Summary

Name Keyword Preview Location Modified Time

Unalloc_4_121344_1474560 address listed below.«George» Montgomery3467 Main : /img_Ch01InChap01.dd/\$Unalloc/Unalloc_4_121344_147 0000-00-00 00:00:00

Client Info.mdb Ballard WA 98107 S Thomas «Geor /img_Ch01InChap01.dd/Client Info.mdb 2005-12-09 06:53:58 V

Billing Letter.doc address listed below.«George» Montgomery3467 Main : /img_Ch01InChap01.dd/Billing Letter.doc 2005-12-09 06:50:28 V

confirmation.txt you for your business.«George»----- /img_Ch01InChap01.dd/confirmation.txt 2005-12-09 06:52:58 V

r0000000_13_October_2003.doc address listed below.«George» Montgomery3467 Main : /img_Ch01InChap01.dd/\$CarvedFiles/1/r0000000_13_Oc 0000-00-00 00:00:00

Income.xls 00 \$ 800.00 Thomas «George» \$ 450.00 /img_Ch01InChap01.dd/Income.xls 2005-12-09 06:52:18 V

r0000049_02_November_2003.doc nowhere.com/Regards.«George» Montgomery----- /img_Ch01InChap01.dd/\$CarvedFiles/1/r0000049_02_N 0000-00-00 00:00:00

letter1.txt Please contact me ASAP.«George»----- /img_Ch01InChap01.dd/letter1.txt 2005-12-09 06:51:50 V

Regrets.doc nowhere.com/Regards.«George» Montgomery----- /img_Ch01InChap01.dd/Regrets.doc 2005-12-09 06:50:52 V

George

Save Table as CSV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

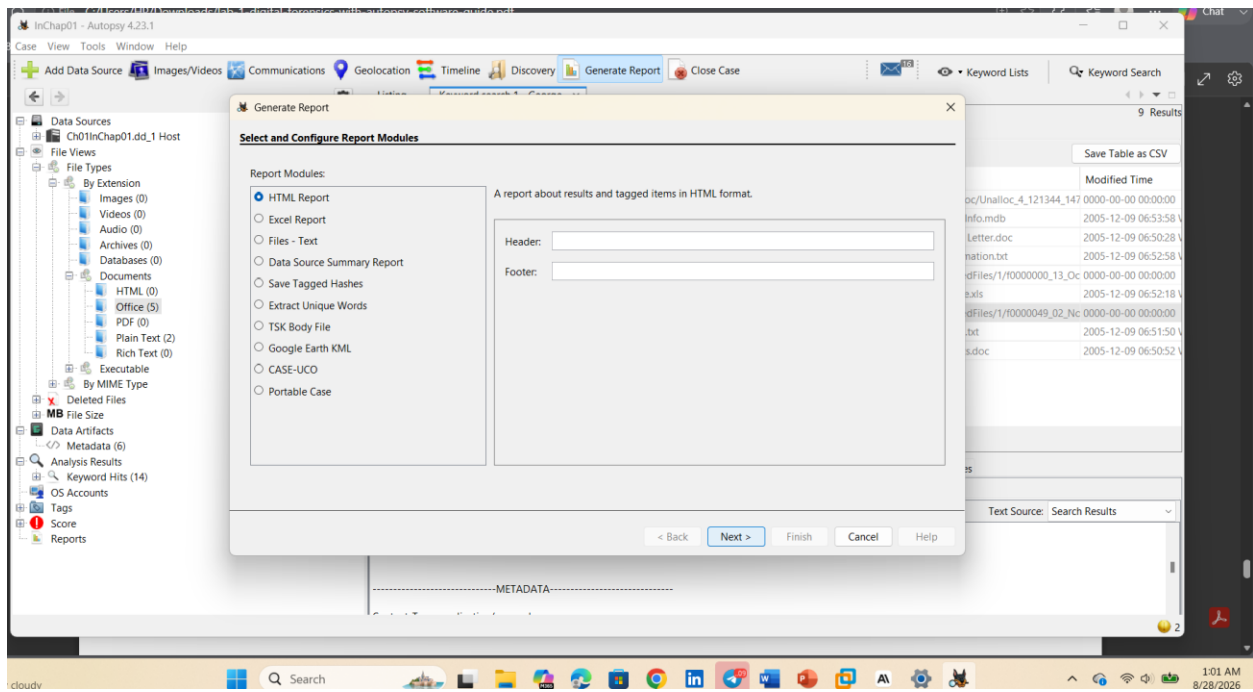
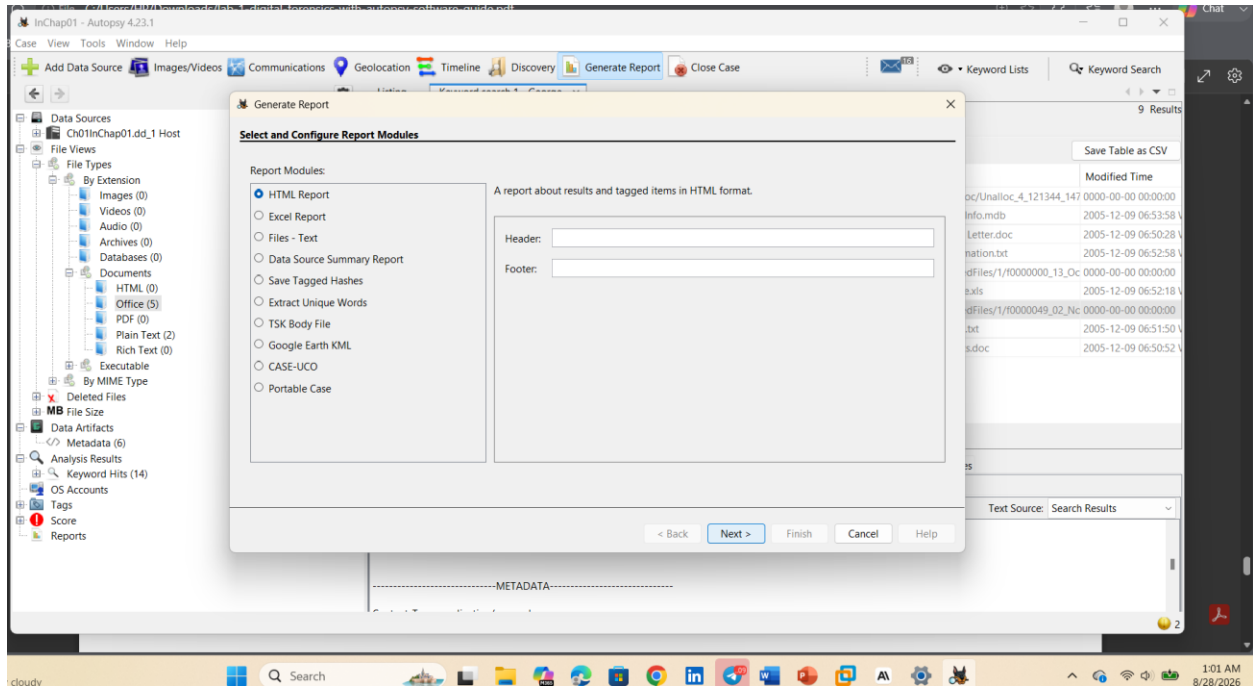
Strings Extracted Text Translation

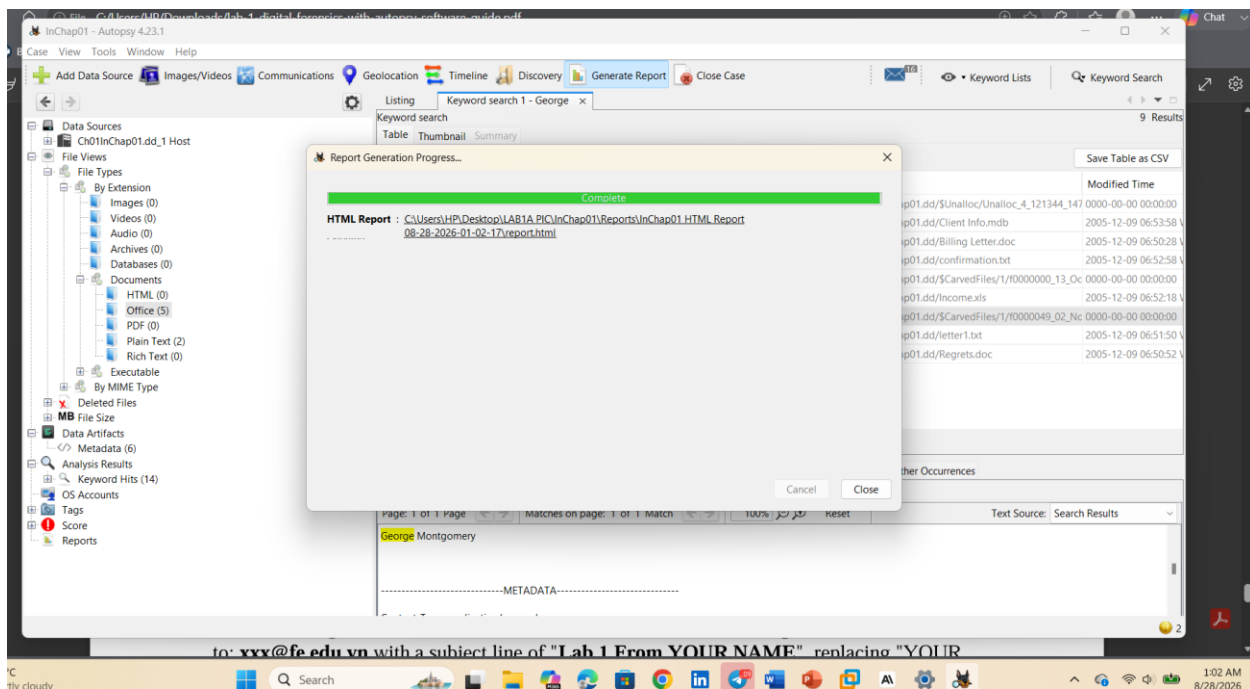
Page: 1 of 1 Page Matches on page: 1 of 1 Match 100% Reset Text Source: Search Results

George Montgomery

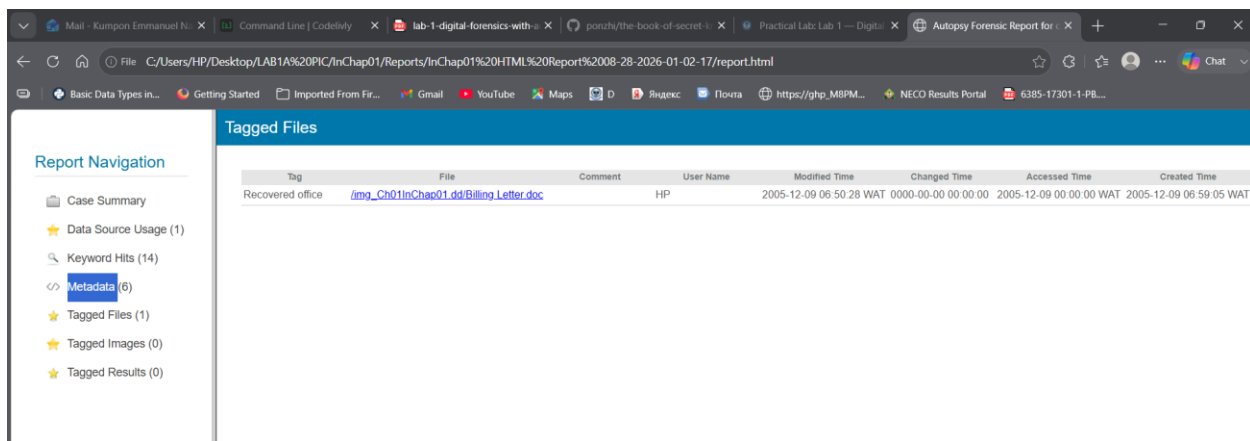
-----METADATA-----

Generate reports



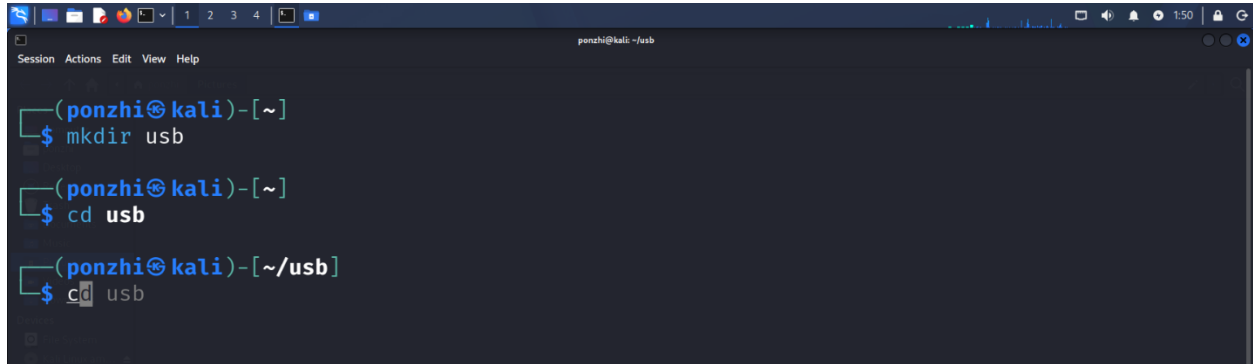


Open or save as file doc recovered



Part B Sleuth Kit Analysis

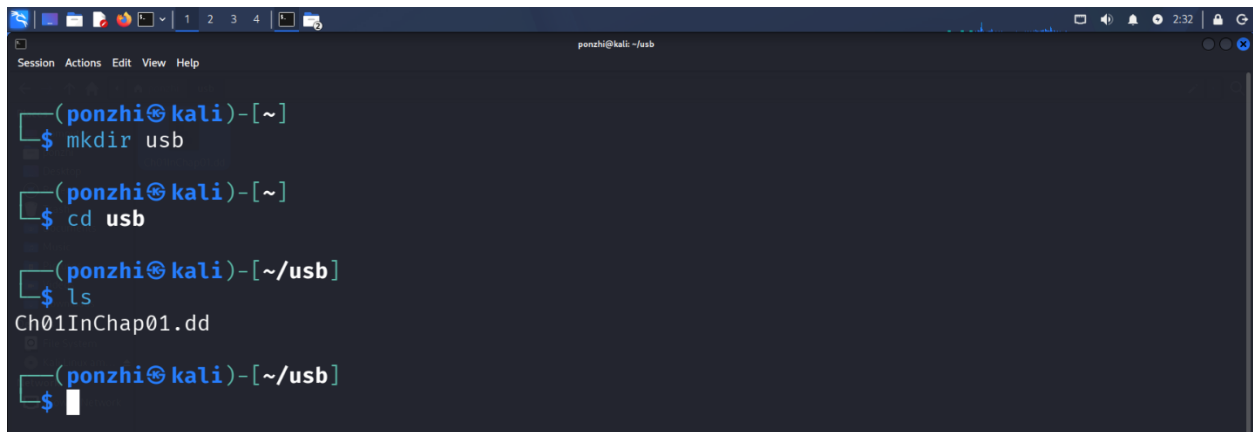
Dedicated Lab folder



```
(ponzhi@kali)~$ mkdir usb
(ponzhi@kali)~$ cd usb
(ponzhi@kali)~/usb$
```

A terminal window titled 'ponzhi@kali: ~/usb' showing the creation of a directory named 'usb' and navigation into it. The prompt changes from '~' to '~/usb' after the 'cd' command.

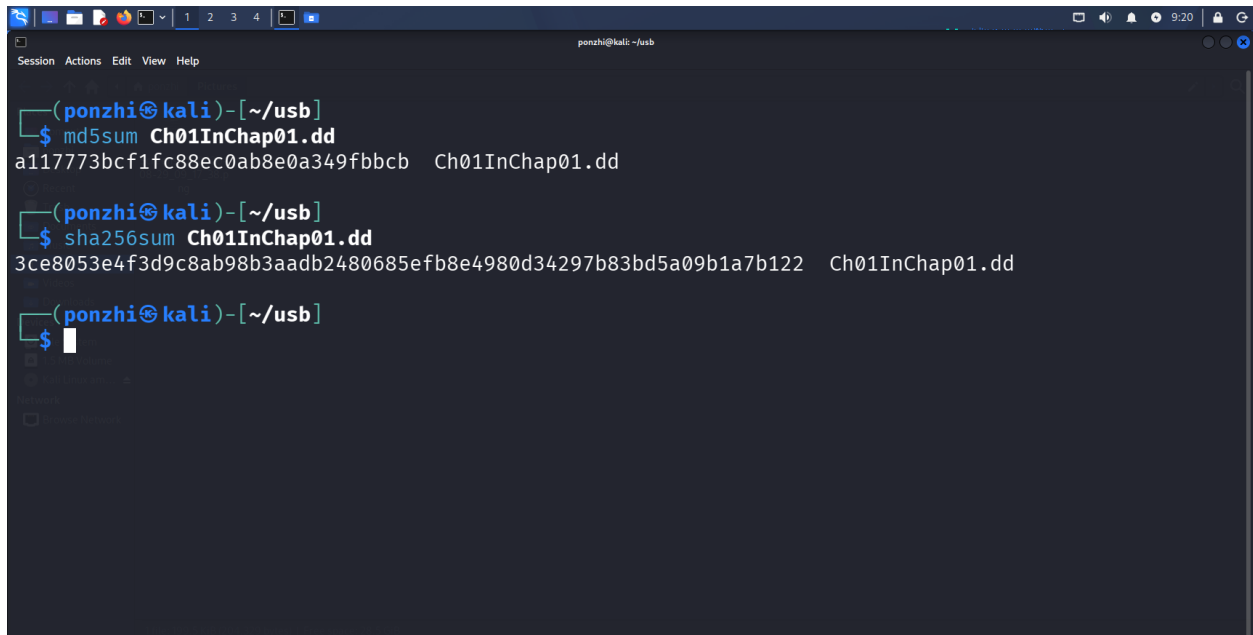
Moving the Forensic image into Usb



```
(ponzhi@kali)~$ mkdir usb
(ponzhi@kali)~$ cd usb
(ponzhi@kali)~/usb$ ls
Ch01InChap01.dd
(ponzhi@kali)~/usb$
```

A terminal window titled 'ponzhi@kali: ~/usb' showing the same directory creation and navigation as the previous screenshot. The 'ls' command is used to list the contents of the 'usb' directory, showing 'Ch01InChap01.dd'.

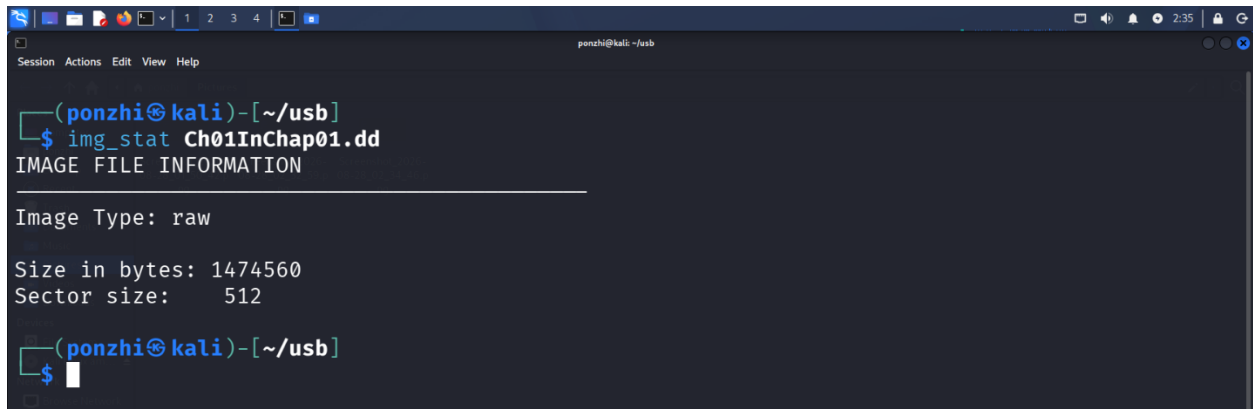
MD5 and SHA-256 hashes of Ch01InChap01



```
(ponzhi@kali)~/usb$ md5sum Ch01InChap01.dd
a117773bcf1fc88ec0ab8e0a349fbccb  Ch01InChap01.dd
(ponzhi@kali)~/usb$ sha256sum Ch01InChap01.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122  Ch01InChap01.dd
(ponzhi@kali)~/usb$
```

A terminal window titled 'ponzhi@kali: ~/usb' showing the calculation of MD5 and SHA-256 hashes for the file 'Ch01InChap01.dd'. The 'md5sum' command outputs 'a117773bcf1fc88ec0ab8e0a349fbccb' and the 'sha256sum' command outputs '3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122'.

Showing the details of the image format



```
(ponzhi@kali)-[~/usb]
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size: 512

(ponzhi@kali)-[~/usb]
$
```

Showing file system details and statistics, including layout, sizes and labels



```
(ponzhi@kali)-[~/usb]
$ fsstat Ch01InChap01.dd
FILE SYSTEM INFORMATION
-----
File System Type: FAT12

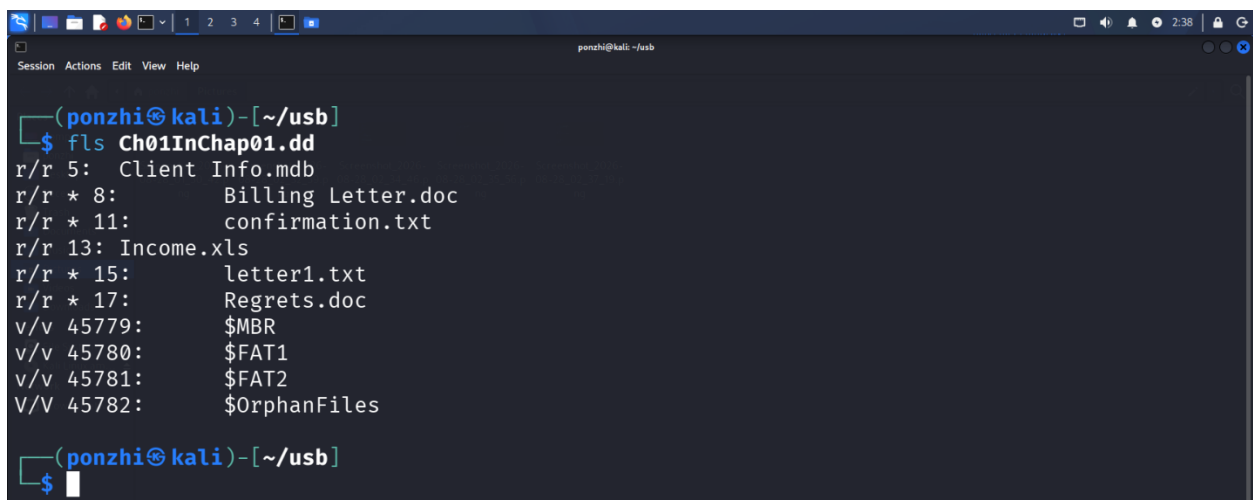
OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: 3MM

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18

(ponzhi@kali)-[~/usb]
$
```

List allocated and deleted file names in a directory



```
(ponzhi@kali)-[~/usb]
$ fls Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles

(ponzhi@kali)-[~/usb]
$
```

-d Display deleted entries only

```
(ponzhi@kali)-[~/usb]
$ fls -d Ch01InChap01.dd
r/r * 8:      Billing Letter.doc
r/r * 11:     confirmation.txt
r/r * 15:     letter1.txt
r/r * 17:     Regrets.doc

(ponzhi@kali)-[~/usb]
$
```

Recover deleted files using inode without specifying a file name

```
(ponzhi@kali)-[~/usb]
$ icat Ch01InChap01.dd 15 > letter1.txt

(ponzhi@kali)-[~/usb]
$ cat letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

(ponzhi@kali)-[~/usb]
$
```

Extract the data units of letter1.txt using inode 15

```
(ponzhi@kali)-[~/usb]
$ icat Ch01InChap01.dd 15
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

(ponzhi@kali)-[~/usb]
$
```

Display the statistics and details about a given meta data structure

```
Session Actions Edit View Help
(ponzhi@kali)-[~/usb]
$ icat Ch01InChap01.dd 15
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

(ponzhi@kali)-[~/usb]
$ istat Ch01InChap01.dd 15
Directory Entry: 15
Not Allocated
File Attributes: File, Archive
Size: 121
Name: _ETTER1.TXT

Directory Entry Times:
Written:      2005-12-09 06:51:50 (WAT)
Accessed:     2005-12-09 00:00:00 (WAT)
Created:      2005-12-09 06:59:09 (WAT)

Sectors:
312
```

Extract the contents of a given data unit.

List the details about units and can extract the unallocated space of the file system.

```
Session Actions Edit View Help
(ponzhi@kali)-[~/usb]
$ blkcat Ch01InChap01.dd 312
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

(ponzhi@kali)-[~/usb]
$ blkls Ch01InChap01.dd 312-312
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

(ponzhi@kali)-[~/usb]
$
```

Extract the contents of a given data unit in hex

```
ponzhi@kali: ~/usb
$ blkcat Ch01InChap01.dd 312 | xxd
00000000: 4561 726c 2c20 0d0a 5765 206e 6565 6420  Earl, ..We need
00000010: 746f 206d 6565 7420 6f6e 2074 6865 2031  to meet on the 1
00000020: 3874 6820 6f66 2041 7567 7573 7420 746f  8th of August to
00000030: 2063 6f6e 6669 726d 2074 6865 2077 6f72  confirm the wor
00000040: 6b20 4920 616d 200d 0a64 6f69 6e67 2066  k I am ..doing f
00000050: 6f72 2079 6f75 2e20 506c 6561 7365 2063  or you. Please c
00000060: 6f6e 7461 6374 206d 6520 4153 4150 2e0d  ontact me ASAP..
00000070: 0a0d 0a47 656f 7267 6500 0000 0000 0000  ... George.....
00000080: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000090: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000a0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000b0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000c0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000d0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000e0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000f0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000100: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000110: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000120: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000130: 0000 0000 0000 0000 0000 0000 0000 0000  .....
```

Showing a file with multiple sectors

```
ponzhi@kali: ~/usb
$ istat Ch01InChap01.dd 13
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written:      2005-12-09 06:52:18 (WAT)
Accessed:     2005-12-09 00:00:00 (WAT)
Created:      2005-12-09 06:59:07 (WAT)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311

ponzhi@kali: ~/usb
$
```

Mounting a USB image



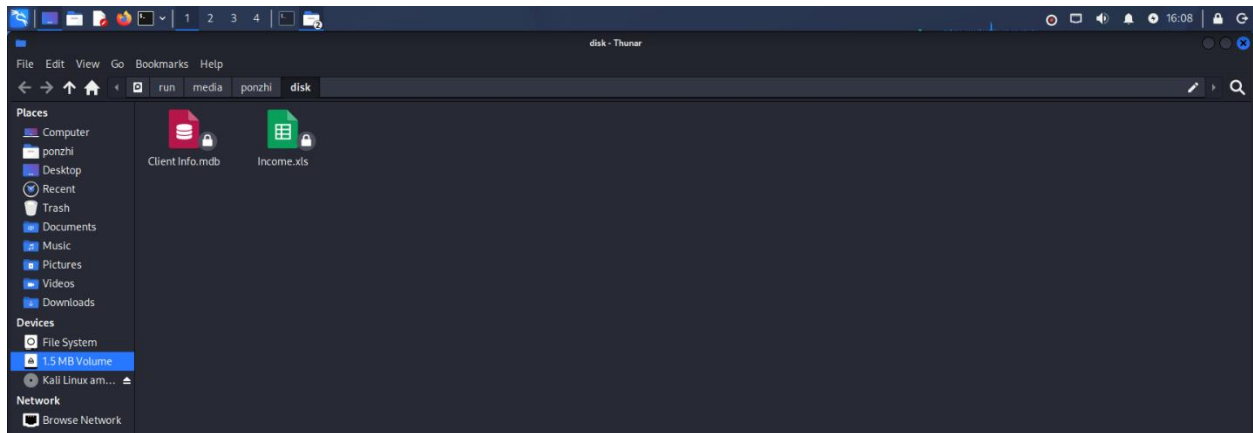
```
(ponzhi@kali)-[~/usb]
$ sudo losetup --partscan --find --show --read-only Ch01InChap01.dd
[sudo] password for ponzhi:
/dev/loop0

(ponzhi@kali)-[~/usb]
$
```

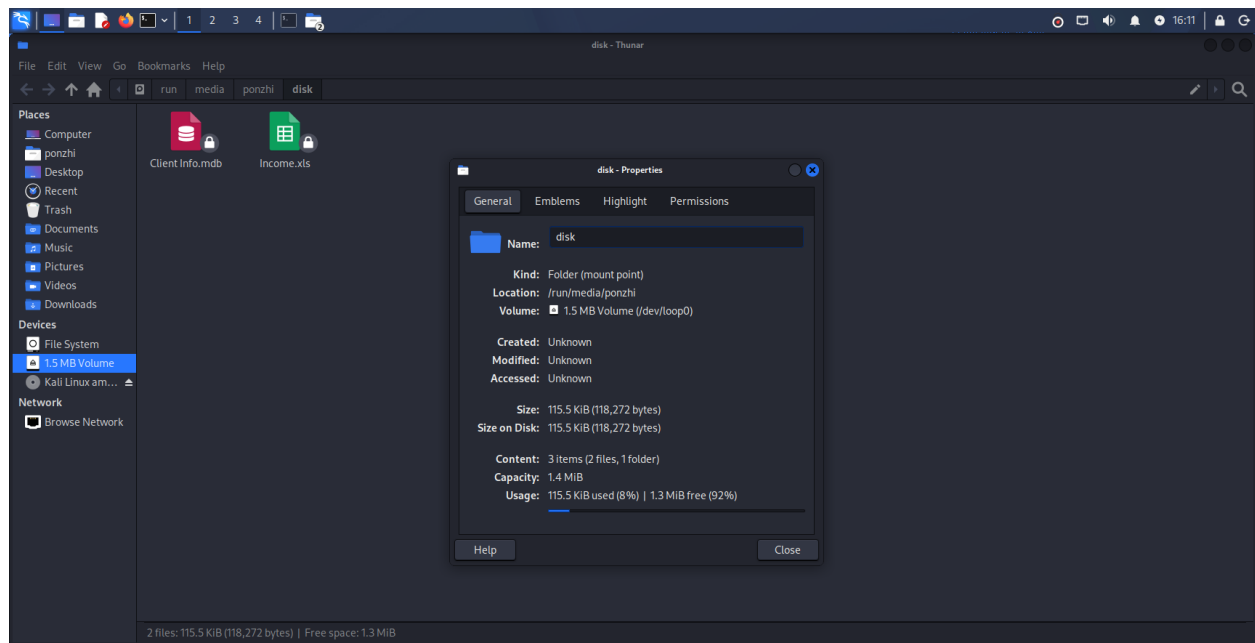
Partscan: This option tells losetup to scan the partition table in the specified disk image (Ch01InChap01.dd) and automatically create loop devices for each partition found. This is useful when dealing with disk images that contain multiple partitions.

Find: This option instructs losetup to find the first available loop device. When combined with --show, it ensures that the command displays the path of the loop device that is being used

Showing Content



Showing the mounted Path



Showing the disk content

```

ponzhi@kali: /run/media/ponzhi/disk
$ ls
Ch01InChap01.dd  letter1.txt

(ponzhi@kali)-[~/usb]
$ cd /run/media/ponzhi

(ponzhi@kali)-[/run/media/ponzhi]
$ ls -l
total 7
drwxr-xr-x 2 ponzhi ponzhi 7168 Jan  1  1970 disk

(ponzhi@kali)-[/run/media/ponzhi]
$ cd /run/media/ponzhi/disk

(ponzhi@kali)-[/run/media/ponzhi/disk]
$ ls -l
total 116
-rw-r--r-- 1 ponzhi ponzhi 104448 Dec  9  2005 'Client Info.mdb'
-rw-r--r-- 1 ponzhi ponzhi 13824 Dec  9  2005 Income.xls

(ponzhi@kali)-[/run/media/ponzhi/disk]
$

```

Hashing the files content

```
ponzhi@kali: /run/media/ponzhi/disk
Session Actions Edit View Help

(ponzhi@kali)-[/run/media/ponzhi]
$ cd /run/media/ponzhi/disk

(ponzhi@kali)-[/run/media/ponzhi/disk]
$ ls -l
total 116
-rw-r--r-- 1 ponzhi ponzhi 104448 Dec  9  2005 'Client Info.mdb'
-rw-r--r-- 1 ponzhi ponzhi 13824 Dec  9  2005 Income.xls

(ponzhi@kali)-[/run/media/ponzhi/disk]
$ md5sum Income.xls "Client Info.mdb"
6a2e65afc5af4fc5f9da2859df134eac Income.xls
eccbad3940b179265dc597b793576770 Client Info.mdb

(ponzhi@kali)-[/run/media/ponzhi/disk]
$ sha256sum Income.xls "Client Info.mdb"
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d Income.xls
e0e66e37079c0e429c7ad8436adc68276a71f2b51be2ccda5392ece9ef416809 Client Info.mdb

(ponzhi@kali)-[/run/media/ponzhi/disk]
$
```

Conclusion

This examination successfully showed how to recover a deleted file from a forensic disk image using both graphical (Autopsy) and command-line (The Sleuth Kit) methods. We found the target file, INCOME.XLS, in the deleted files section of the Ch01InChap01.dd image using Autopsy's Deleted Files feature. We confirmed this finding through Sleuth Kit's fls command, which also showed the file's metadata and inode address.

We tried three different methods to recover the file: icat and blkcat. After recovery, we opened the spreadsheet in Excel, which confirmed that it was complete and readable. This step proved that we recovered not just raw data but usable information.

Throughout this analysis, we ensured the original evidence image remained unchanged. We recorded the MD5 and SHA-256 hash values for Ch01InChap01.dd before starting our work, and they stayed the same, following proper procedures to maintain the chain of custody.

This exercise highlighted the importance of checking results with different forensic tools and confirming that recovered evidence is both intact and practical to use, beyond just matching hash values.